

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2919 ±0.00529	0.6592 ±0.0103	0.4572 ±0.0221	0.3311 ±0.00441
Cramér $\ell_2^2 \downarrow$	0.03014 ±0.00135	0.04859 ±0.00223	0.05131 ±0.00275	0.03119 ±0.00142
CF-$\ell_2^2/\omega^2 \downarrow$	0.02403 ±0.000231	0.03875 ±0.00146	0.04395 ±0.00373	0.02542 ±0.000502

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2213 ±0.0166	0.5569 ±0.0251	0.4234 ±0.0252	0.2565 ±0.0161
Cramér $\ell_2^2 \downarrow$	0.01302 ±0.00198	0.03549 ±0.00292	0.0484 ±0.00534	0.01446 ±0.00234
CF-$\ell_2^2/\omega^2 \downarrow$	0.01257 ±0.00191	0.03166 ±0.00288	0.04733 ±0.00524	0.01382 ±0.00206

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1429 ±0.0184	0.4831 ±0.0634	0.2513 ±0.0626	0.1788 ±0.0176
Cramér $\ell_2^2 \downarrow$	0.005543 ±0.00187	0.01837 ±0.00322	0.01233 ±0.00368	0.005541 ±0.00164
CF-$\ell_2^2/\omega^2 \downarrow$	0.004798 ±0.00187	0.01535 ±0.00281	0.01105 ±0.00335	0.004844 ±0.00161

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2898 ±0.00628	0.4937 ±0.0459	0.4034 ±0.0515	0.3181 ±0.00934
Cramér $\ell_2^2 \downarrow$	0.02811 ±0.00161	0.0314 ±0.00732	0.03939 ±0.00341	0.02893 ±0.00142
CF-$\ell_2^2/\omega^2 \downarrow$	0.02235 ±0.00152	0.02601 ±0.00537	0.03415 ±0.0038	0.02284 ±0.000706

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1517 ±0.0213	0.3521 ±0.0356	0.204 ±0.0275	0.167 ±0.0166
Cramér $\ell_2^2 \downarrow$	0.005731 ±0.00157	0.009318 ±0.00226	0.008978 ±0.00191	0.004332 ±0.00119
CF-$\ell_2^2/\omega^2 \downarrow$	0.005105 ±0.00143	0.008819 ±0.00253	0.00851 ±0.00186	0.003754 ±0.00108